

Osteoporosis is a degenerative bone condition in which healthy bone degenerates into porous bone. Because bone provides the structural integrity that allows the musculoskeletal system to bear weight, osteoporosis is associated with a higher risk of bone fracture.

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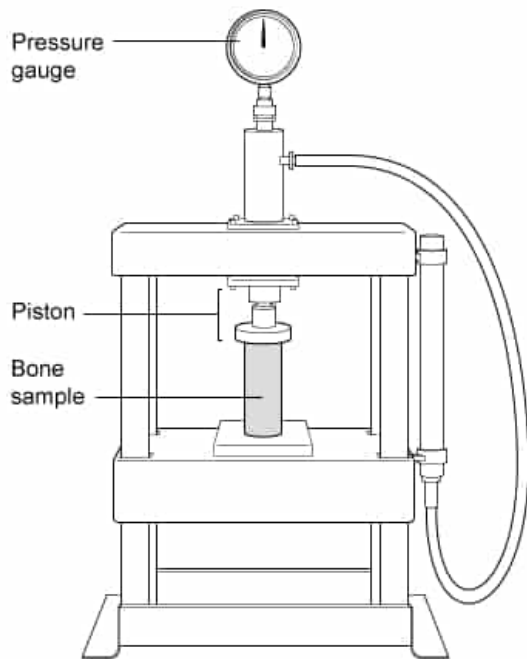


Figure 1 Bone compression device loaded with a bone sample

The force required to compress the bone samples longitudinally (lengthwise) by a given distance was measured, with the results shown in Figure 2.

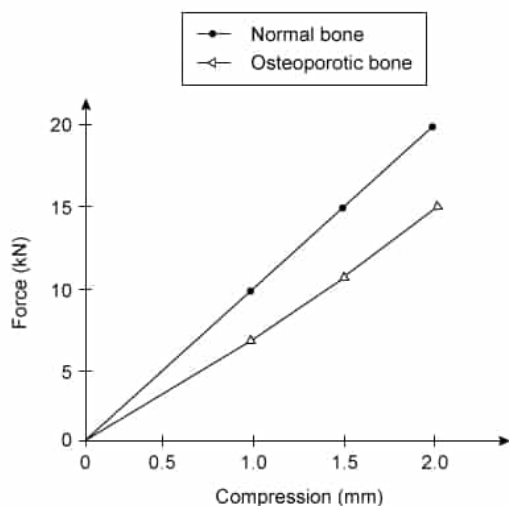


Figure 2 Relationship between force and compression for normal and osteoporotic bones

Researchers observed that bone samples returned to their original dimensions for deformations not exceeding 1% of the original length L_0 . For each deformation, the

associated strain ϵ is defined as the change in length of an object ΔL divided by the object's original length (Equation 1).

$$\epsilon = \frac{\Delta L}{L_0}$$

Equation 1

Based on these observations, researchers concluded that both osteoporotic bone and normal bone behaved as an elastic material under a wide range of applied forces.

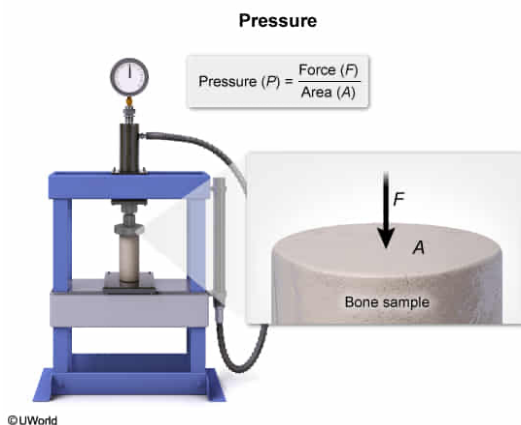
What is the pressure required to compress a normal bone sample by 1.5 mm? (Note: The cross-sectional area of bone samples used in the experiment is 10 mm².)

- ☐ A. 1.10 kN/mm² [4%]
✓ ☒ B. 1.50 kN/mm² [85%]
☐ C. 110 kN/mm² [2%]
☐ D. 150 kN/mm² [7%]

Correct

86% answered correctly

Explanation:



Pressure (P) describes a **force (F)** that is **applied perpendicularly** to a surface with **area (A)**. As such, the magnitude of pressure is equal to force per unit area:

$$P = \frac{F}{A}$$

In simple mechanical systems, pressure is often caused by forceful contact between two solid objects. Consequently, the pressure experienced by any object within a two-object system is **directly proportional** to the magnitude of force between the objects and **inversely proportional** to the area over which the contacting objects meet.

As seen from **Figure 2**, compressing a normal bone sample by 1.5 mm requires 15 kN of force. Furthermore, the question states that the bone sample has a uniform cross-sectional area of 10 mm². According to Figure 1, the cross-sectional area of the bone sample is smaller

than the cross-sectional area of the compressive piston such that the cross-sectional area of the bone sample is the area of contact between compressive piston and bone.

Consequently, the pressure exerted on the bone is calculated as:

$$P = \frac{15 \text{ kN}}{10 \text{ mm}^2} = 1.5 \frac{\text{kN}}{\text{mm}^2}$$

Therefore, using a compressive device to compress a normal bone sample by 1.5 mm exerts a pressure of 1.5 kN/mm² onto the bone sample **(Choice B)**.

Educational objective:

The pressure exerted on a surface is the force applied perpendicular to the surface divided by the area of the surface. In mechanical systems, the surface area onto which the pressure is exerted is usually equivalent to the area of contact between two objects.

Time spent:0 sec

Subject:
Physics

QID:400295

System:
Mechanics and Energy

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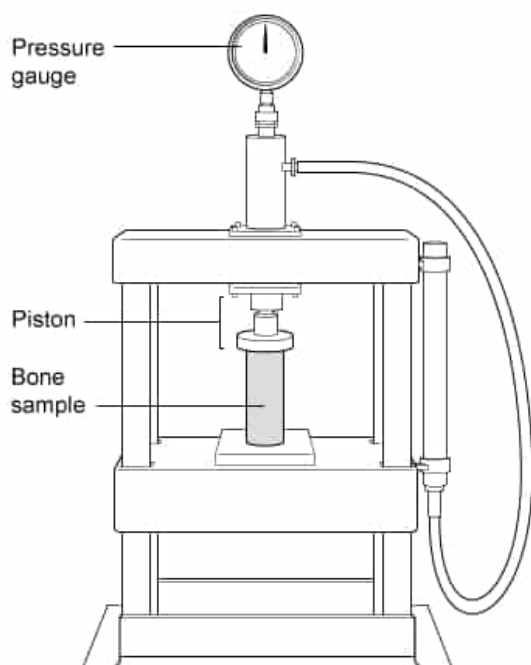


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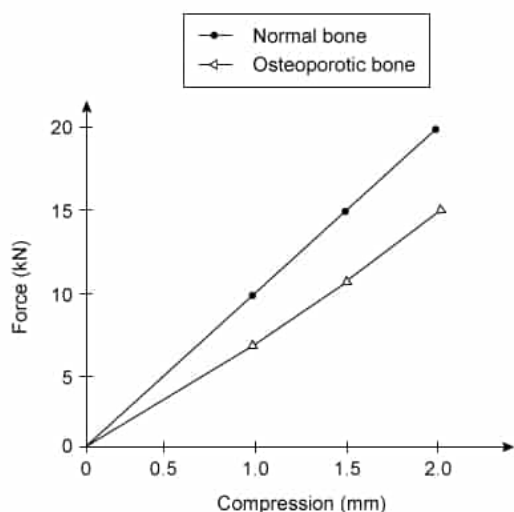


Figure 2 Relationship between force and compression for normal and osteoporotic bones

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associated strain ϵ is defined as the change in length of an object ΔL divided by the object's original length (Equation 1).

$$\epsilon = \frac{\Delta L}{L_0}$$

Equation 1

Based on these observations, researchers concluded that both osteoporotic bone and normal bone behaved as an elastic material under a wide range of applied forces.

Compared to osteoporotic bone, the work required to compress normal bone by 2 mm is:

- ☐ A. 3/4 as great. [9%]
- ☐ B. the same.
- ☒ C. 4/3 times greater. [88%]
- ☐ D. 8/3 times greater. [1%]

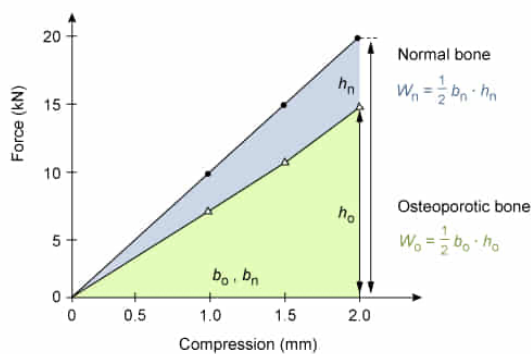
Correct

87% answered correctly

Explanation:

Work from area under the curve
(force vs compression)

Work (W) = Force (F) · Distance (D)
 W = area under the curve
 $W = A_T$



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Mechanical work describes the transfer of energy that occurs through the application of a **force (F)** to move an object over some **distance (d)** or the application of pressure (P) over some change in volume (ΔV):

$$W = F \cdot d = P \cdot \Delta V$$

Consequently, in a graph of force versus distance, the **area under the curve** is equal to the associated work because within the above equation, F corresponds to a y-axis value (a force) and d corresponds to an x-axis value (a displacement from the equilibrium point).

Figure 2 shows that the application of a compressive force results in the displacement of the samples of both bone types away from their equilibrium positions (ie, original, noncompressed

length). As such, the compressing machine performs mechanical work as the surface of the bone is displaced some distance from its equilibrium position.

The corresponding work performed over 2 mm of compression can be visualized graphically as the area A_T occupied by a triangle with a base b that is 2 mm (0.002 m) in length and a height h that is equal to the applied change in force:

$$W = A_T = \frac{1}{2}b \cdot h$$

Applying this relationship, the work performed on the normal bone (W_n) and on the osteoporotic bone (W_o) may be calculated as follows:

$$W_n = \frac{1}{2}b_n h_n = \frac{1}{2} \cdot (0.002 \text{ m}) \cdot (20,000 \text{ N}) = 20 \text{ J}$$

$$W_o = \frac{1}{2}b_o h_o = \frac{1}{2} \cdot (0.002 \text{ m}) \cdot (15,000 \text{ N}) = 15 \text{ J}$$

Accordingly, the ratio between W_n and W_o is:

$$\frac{W_n}{W_o} = \frac{20 \text{ N} \cdot \text{m}}{15 \text{ N} \cdot \text{m}} = \frac{4 \cdot \cancel{5 \text{ N} \cdot \text{m}}}{3 \cdot \cancel{5 \text{ N} \cdot \text{m}}} = \frac{4}{3}$$

Therefore, it can be concluded that the work needed to compress normal bone by 2 mm is 4/3 times greater than the work needed to compress osteoporotic bone by 2 mm (**Choice C**).

Educational objective:

Work is an energy transfer that takes place due to a force acting over some distance. In a graph of the force acting on the object versus the distance the object is displaced, the corresponding work performed on the object is equal to the area under the curve.

Time spent:0 sec

Subject:
Physics

QID:400296

System:
Mechanics and Energy

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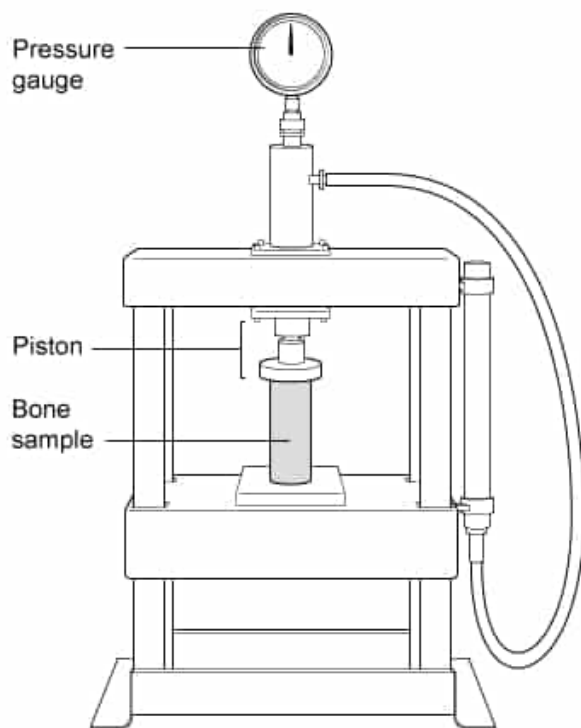


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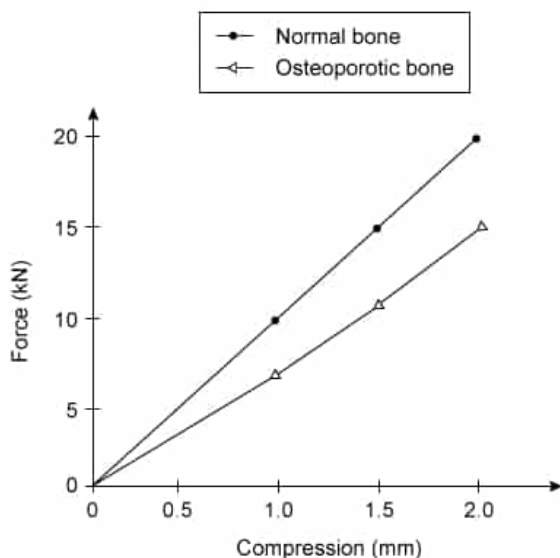


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$$\varepsilon = \frac{\Delta L}{L_0}$$

Equation 1

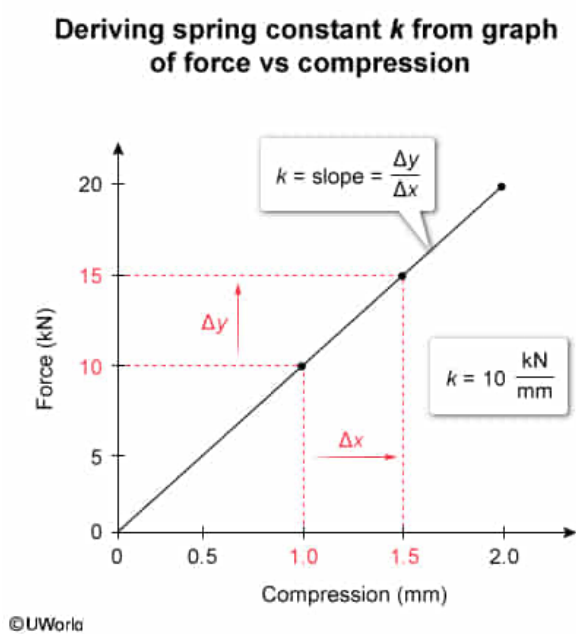
Based on these observations, researchers concluded that both osteoporotic bone and normal bone behaved as an elastic material under a wide range of applied forces.

According to the data in Figure 2, what is the spring constant k for normal bone?

- ☐ A. 7.5 kN/mm [2%]
- ✓ ☒ B. 10 kN/mm [92%]
- ☐ C. 30 kN/mm [2%]
- ☐ D. 40 kN/mm [1%]

Correct
92% answered correctly

Explanation:



Displacing an elastic object from its **equilibrium point** (ie, its position at rest) generates an elastic force (F_{el}) that counteracts the force displacing the object. The elastic force acts to restore the elastic object to its original position if other forces acting on the object are removed.

Hooke's law describes the elastic forces of perfectly elastic materials. It states that F_{el} is a function of object displacement (x) and the material's spring constant (k), which is a measure of the relative elasticity (stiffness) of a spring-like object:

$$F_{el} = -kx$$

In practice, most elastic objects are imperfectly elastic, and minimally or maximally displacing an imperfectly elastic, spring-like object from its equilibrium point generates an elastic force greater or less than that predicted by Hooke's law. As a result, Hooke's law must be applied cautiously to elastic materials to avoid error.

In the case of the bone samples described in the passage, **Figure 2** shows that the force needed to compress the bone samples increases in **direct proportion** to the amount of bone compression. Therefore, bone acts as a perfectly elastic, spring-like substance for the displacements observed within the experiment. Accordingly, Hooke's law is well suited to describe the bone samples.

Because the force needed to compress a bone sample (F) is equal and opposite the elastic force resisting compression (F_{el}), the slope of each line within Figure 2 may be used to calculate the spring constant of each type of bone. Evaluating the spring constant for normal bone gives:

$$\begin{aligned} F &= kx \\ k &= \frac{F}{x} = \text{slope} = \frac{(F_2 - F_1)}{(x_2 - x_1)} \\ k &= \text{slope} = \frac{(20 \text{ kN} - 10 \text{ kN})}{(2 \text{ mm} - 1 \text{ mm})} = 10 \frac{\text{kN}}{\text{mm}} \end{aligned}$$

Therefore, the spring constant of normal bone is 10 kN/mm, which indicates that normal bone requires 10 kN of additional force for each millimeter that the bone is compressed (**Choice B**).

Educational objective:

Hooke's law states that the elastic force generated by an object is a product of the object's spring constant (elasticity) and the extent to which the object is displaced. For perfectly elastic objects, the relationship between displacement and elastic force is linear across the range of displacements.